

There Are No Metaphysical Telescopes in Futures Studies

Abstract

Futures studies often works as if it had some metaphysical telescope: as if futures were out there, like stars, waiting to be viewed, and as if the right methods could bring them into view. Nobody claims to own such a telescope, yet the field acts, speaks and teaches as if one could be had. This paper argues that there is none. Nothing travels to us from the future, so there is no access to futures; and looking changes what is there, so there is no neutral observer of them. Knowledge of futures comes from the past used well and from the present situation, or it does not come at all. The paper then shows where the as-if telescope shapes practice: in questions about what 2050 will look like, in horizons set by round years, in a modal vocabulary that mixes what the world allows with what we feel, in the futures cone, in scenarios that are pictures rather than structured sets, in wild cards, in anticipation and in radical scepticism, and in backcasting, visions, imagination methods and workshops. For each, it asks what an honest version would look like. It closes with a practical proposal, a limits section in every study that makes claims about futures, and with a self-description for the field: futures studies makes explicit, and tests, the assumptions behind claims about futures, and says what can come of the present situation, on what grounds, for how long, and where knowing stops.

Keywords: *futures studies; epistemology; modality; scenarios; futures cone; anticipation*

1. Introduction

There is a picture of possibility that is hard to resist. A possible world is imagined as a distant country, and we, as observers, look at it through a powerful telescope. Saul Kripke, in *Naming and Necessity* (1980, p. 44), dismissed the picture with a dry remark: possible worlds are not discovered by powerful telescopes. The idea I take from the remark reaches far beyond the philosophy of modality. A picture can be wrong without anybody believing it. It works quietly, by deciding which questions seem natural, and those questions look deep only because the picture is taken for granted. I take the idea, not Kripke's own theory of possible worlds; nothing below depends on it.

Now picture a workshop. "Close your eyes," says the facilitator. "It is 2050. Look around. What do you see?" The participants see a great deal, and some of it is even interesting. But nobody is looking at 2050. There is nothing to look at from here, and nothing to look with.

Futures studies, I will argue, often works as if it had some metaphysical telescope. Nobody claims to own one. Ask futures studies researchers whether they can see the future and they will rightly laugh. Then they show you 2040. People act, speak and teach as if a telescope could be had, given enough methods and materials to find futures to watch. And the picture breeds its own questions, the most familiar being: what will the world look like in 2050? Do not ask that. Ask what can come of the present situation, on what grounds, and for how long.

The telescope is metaphysical in two senses.

(i) It would need a metaphysical object: futures out there, like stars, waiting to be viewed.

The many futures of futures studies are not out there.

(ii) It would need to work beyond physics. Nothing travels to us from the future, and no instrument within physics could reach it.

My claim, however, is epistemic. Even if the future exists, it cannot be viewed.

Let me be clear about what I am not saying. I am not saying the field lacks good research; it does not. Nor am I a sceptic: there is plenty of evidence bearing on futures, in plans, in behaviour, in what people bet on. My claim is narrower, and I think it is worse news. The field is confused about, and not humble enough about, what it can fundamentally get.

The complaint that futures studies claims too much is not new, and the field has its answer ready: it does not predict, it explores alternative futures. That answer, as I will show, gives up the crystal ball and keeps the telescope. Philosophers have asked what futures studies can know (Poli, 2017; Virmajoki, 2026a, 2026b). What I add is a diagnosis. The as-if telescope is not a belief anybody holds. It is a habit that shows in practice: in reports that present pictures of 2040 as findings, in formats read as windows, in programmes that claim to found the field while assuming what it should deliver.

I take these up in turn. Section 2 asks where knowledge of futures comes from: the past used well, or nothing. Section 3 turns to starting points: the question asked, the horizon, the present situation. Section 4 asks what holds, and for how long. Section 5 untangles the modal

vocabulary. Section 6 discusses scenarios, surprises and pictures. Section 7 turns to wishes, workshops and imagination. Section 8 asks what the field is when it is honest with itself.

Some of my examples are imagined, and deliberately so. Naming one scenario study as an instance of a practice that many share would be unfair to it, and kind to the others that stay under the radar. Where I name texts, I name them because of their influence: they teach the practice.

A word on scope. What I discuss critically is the Western tradition of futures studies, and we in it need to keep our own nest clean. Other traditions relate to futures in their own ways (Sardar, 1993), and they may well do better. I leave that open.

The argument that follows is deliberately severe. That severity is not meant as mockery, nor as a dismissal of the field or the people who work in it. The point is to challenge familiar assumptions as directly as possible, and to invite readers who are willing to take the challenge to answer it at full strength.

2. Where knowledge of futures comes from

If we cannot look, how do we know anything about futures at all? My answer is unexciting. We know about futures what we can work out from the past and the present, and nothing else. This section argues for that answer, shows two ways of pretending otherwise, and closes with a second reason why no telescope could work even if we had one.

(A) *Traces run one way.* The present is full of traces of the past: documents, ruins, isotopes, tree rings. The historical sciences live on this. A single past event leaves many traces, so a

well-chosen trace, a smoking gun, can settle a question about it (Cleland, 2002). Nothing comparable runs the other way. No future event has left a trace in the present. A plan is not a trace of the future but one of its causes. Traces are effects of what they are traces of. The asymmetry is old (Reichenbach, 1956) and it is not going away.

Real telescopes, in fact, look backwards. The light that reaches the astronomer left its star long ago, and some of the stars we see are no longer there. A telescope for futures would need light from things that have not yet happened.

Do we see through a microscope? Hacking (1981) thought the question worth asking, because there is something on the other side of the lens. For futures, the question cannot even be asked.

Futures are, in one sense, observable. Van Fraassen (1980) counts the moons of Jupiter as observable because, under suitable circumstances, we could go and look. Futures are observable in the same sense: wait, and they arrive. That is the only way to observe them.

(B) *Gibson's plane ticket*. "The future is already here; it's just not very evenly distributed."

The line is William Gibson's, though its first utterance is not securely dated, and the field likes to read it as method: go where the future already is, and look. In my terms, Gibson replaces the telescope with a plane ticket. I concede what is true in it. Things do spread, and studying the places where they start is serious research (Rogers, 2003). But what one finds there is the present, somewhere else. It is a seed, not a sighting. Whether it spreads depends on what holds: costs, rules, uptake, rivals. Plenty of futures that were "already here" never

arrived elsewhere; Microsoft Bob and Nintendo's Virtual Boy are the polite examples. A plane ticket takes you somewhere else, not somewhen else.

(C) *The dilemma*. If there is no sight, knowledge of futures must come from what we already know, about how things work and about the situation we are in. We know a great deal of both, and we act on it all the time, mostly with success. When we fail, we fail in one of two ways.

(i) We were wrong about how things work.

(ii) We were wrong about the situation.

Some shocks are both. A pandemic first changes the situation, and then rewrites parts of how things work: where people work, how they travel, whom they trust. Where, then, does knowledge of how things work come from? From the past, but not naively. Extending a trend line is not using the past well. Explaining why past episodes went as they did is, because explanation yields mechanisms that may recur, not outcomes that repeat (Virmajoki, 2023). Hence a dilemma. Either the past, used well, gives us knowledge that bears on futures, or there is no futures studies. Unless, of course, someone builds a metaphysical telescope.

Someone might object that this makes futures studies a branch of history. It does not. History asks what happened and why. Futures studies asks what can come of the present situation, on what grounds, and for how long. The questions differ; the evidence does not. And the evidence has a ceiling. The course of history depends on the growth of knowledge, and we

cannot predict what we will come to know (Popper, 1957). This is not scepticism. It is the humility the field keeps promising in its first sentence and forgetting in its second.

(D) *The telescope inside the system*. One programme claims to put all this on a new foundation. Miller, Poli and Rossel (2018) present the discipline of anticipation as the foundations of futures literacy, and argue that it is a discipline because it has a territory of knowledge of its own. The notion of anticipation this literature builds on goes back to Rosen's definition of an anticipatory system: "a system containing a predictive model of itself and/or of its environment, which allows it to change state at an instant in accord with the model's predictions pertaining to a later instant" (Rosen, 2012, p. 313). Studying such systems is perfectly fine. Making them a foundation is not, for three reasons.

(i) Circularity. "Predictive" does all the work in the definition. The definition presupposes knowledge of futures; it does not explain where that knowledge comes from. A foundation that assumes what the building is for founds nothing.

(ii) The homunculus. Explaining our relation to futures by placing a predictive model inside every system is the old homunculus move. A little telescope now sits inside the system, and the question of how it sees starts again inside.

(iii) The heater. A smart thermostat that reads tomorrow's weather forecast is an anticipatory system in Rosen's sense. A concept that generous is too cheap to found a discipline.

The territory of knowledge, then, is borrowed. It comes from the forecasts and foresight it presupposes. There is a way out, and it is instructive. Homunculi are harmless, Dennett

(1978) argued, if they are discharged, that is, explained by teams of dumber parts until nothing clever is left. Anticipation studies never discharges its telescope. Research is precisely the discharging: replacing the little telescope with what we know about how things work, with data, and with stated assumptions about what will keep holding.

(E) *The observer*. Suppose, against everything above, that a telescope could be had. It would still fail, because looking changes what is there. Telescopes do not move stars. Forecasts move markets. After the Black–Scholes model was adopted, option prices came to fit the model better than before; after 1987, they stopped (MacKenzie, 2006). This is performativity, and it is neither rare nor mysterious. Under the right conditions a published picture of the future becomes one of its causes, a self-fulfilling or self-defeating prophecy (Merton, 1948; Popper, 1957, on the “Oedipus effect”). So there are two failures, not one. There is no access to futures, and there is no neutral observer of them. The rest of this paper returns to them wherever they show.

3. Starting points

Every study of futures starts somewhere. It starts with a question, with a horizon, and with a description of where we are. The as-if telescope quietly shapes all three, and each can be set right without any new method, only with more care about what is being asked and from where.

(A) *The question*. “What will the world look like in 2050?” The question invites a description of a place, and it gets one. Questions have presuppositions, and a question whose presupposition is false has no true direct answer (Belnap & Steel, 1976). Collingwood made

the point for historical inquiry long ago: what we find depends on what we ask, and what we ask depends on what we take for granted (Collingwood, 1939, 1940). The question about 2050 takes for granted that 2050 is there to be described, as if it were a city one has visited. That presupposition is false, and so every confident answer to the question is an answer to a different question, one that nobody asked aloud. What can be asked instead is this: what can come of the present situation, on what grounds, and for how long? The difference is not pedantry. The first question can be answered by anybody with confidence and a word processor. The second can be answered only by someone who knows something, and it can be answered badly in ways that others can check.

The field has its own typology of questions. Börjeson et al. (2006) sort scenario work by the question it answers.

(i) What will happen? Predictive scenarios.

(ii) What can happen? Explorative scenarios.

(iii) How can a specific target be reached? Normative scenarios.

As a map of questions, the typology is useful. As usually read, however, it is as if the typology offered three telescopes from the same supermarket: one for the likely future, one for the possible futures, one for the desired one. Each question is posed about “the future” as an object. Each can be reformulated so that it is posed about the present situation and its grounds, and only then does it become a research question.

Someone might object that the question about 2050 is not ours to choose. Those who commission futures research ask it, and they pay for an answer. But making those who pay happy is one thing, and serious research is another. Consider a physician. What makes a physician a physician is not that the patients are satisfied, or even that many of them recover, but that the physician's judgement rests on what medicine knows: tests, treatments, what usually follows what. A healer without that knowledge is no physician, however many patients leave happy. The futures studies researcher works as if in the physician's position, with one difference: fewer base rates. Asked "Will I be fine?", the physician says what the tests show, what usually follows, and what would change the picture. Asked "What will 2050 look like?", the futures studies researcher can say what can come of these developments by around 2050, given what, and how sure we can be. The commissioner loses a picture and gains something to act on.

(B) *The horizon*. Studies of futures love round years: 2030, 2040, 2050. The round year is a symptom. It is chosen as a viewing point, as if one could set the telescope to a distance, and not because anything we know runs out there. Why is five years from now a future while five minutes is not? The question is not a joke. It shows that horizons are usually set by convention, and a convention can hide a picture. A horizon should come from two other places.

(i) From what we know. Different kinds of knowledge reach different distances. Even where the physics is known, the reach is finite: in Lorenz's model, the largest atmospheric scales remain predictable only a few weeks in advance (Lorenz, 1969). The meteorologist knows this and says so. Beyond that limit, the meteorologist does not draw pictures of the weather, but says "I do not know", or speaks of climate, which is a different question. Futures studies

speaks of more general things than the weather, and nobody expects of it the forecaster's accuracy, so the lesson is not that after ten days we should say nothing. The lesson is the analogy: we must admit where knowledge stops and speculation starts. The line is not always sharp, but it is there. The number of people who will be over 80 in 2045 can be stated with some confidence, because they are all alive now. Next year's oil price cannot. One should not fix a temporal scale and then ask what is possible within it. The reach of what we know should set the scale. The reader may ask themselves whether they would dare to write down how the world is in a hundred years, so that their great-grandchildren would know where they are.

(ii) From what is needed. Actors and systems have time-needs (Virmajoki & Laakkonen, 2024). A railway, once built, runs for decades, and those who plan it must think in decades whether they like it or not. The time-need sets how far we must think, and what we know sets how far we can. When the two come apart, and they often do, the honest move is to say so and to state conditionals beyond the point where knowledge runs out. The dishonest move is to fill the gap with a picture. The pressure to fill it is real, but filling it is still a choice.

Time horizons in futures research vary with the task (Brier, 2005). A fifteen-year horizon, where used, is at best a heuristic: long enough for things to change, short enough not to become science fiction. As a heuristic it is respectable. As a ritual it works as if it were the telescope's focal length. Precision follows the same logic. A claim about one particular year twenty years from now is odd, while a decade, or a five-year window, is at least honest about what could be known. And the heuristic has exceptions in both directions. Some things stay sharp for decades, as age structures do, until something like a pandemic changes the situation and then rewrites part of how things work.

Someone might object that commissioners need a year, because budgets and strategies are dated. Then give them the year as a horizon of concern, not as a viewing point. “By around 2040” marks what the decision needs. It does not promise a view of 2040.

(C) *The present situation.* Every claim about futures starts from where we are. I call this the present situation, and I mean a period, not an instant: the state of things for which evidence exists. The number of people now over 64 belongs to it, and next year’s election result does not. We need to notice two things about it.

(i) Evidence about the present is always already a little past. Statistics arrive late and are revised, and surveys describe the months in which they were made. Economists even have a word for estimating the present: nowcasting (Giannone, Reichlin & Small, 2008). The trace asymmetry of Section 2 reaches into the present itself. What we call the present situation is, strictly, the recent past seen from a little later.

(ii) The present situation is contested. What AI can do now, how many people actually work remotely, what state the climate system is in: all of these are disputed, and some of the data that describe them are themselves products of models, as reanalysis data in climate science are (Parker, 2017). Taken alone, a contested present is nothing special. What is special is its combination with modality, many domains and reflexivity (Virmajoki, 2026a).

Every science has to establish its present, and does. Elsewhere, this causes no trouble. Almost every research plan opens with “Recently...” and a survey of current work, and nobody asks where the present begins. In futures studies, the present is often skipped on the way to 2050,

as if the view from there would make up for not knowing where one stands. It does not. An error about the situation travels into every claim built on it, and no picture of 2050 corrects it.

(D) *Mega as method*. A trend is respectable: a present regularity, assumed to continue for a while. Megatrends have been with us since Naisbitt (1982), and they now have institutional homes, such as the European Commission's Megatrends Hub (European Commission, 2025). A megatrend works as if it were a trend with a marketing department. The prefix carries the content. It says that the direction is large and the destination known. It does not say why the regularity should continue, for how long, or what would break it. Consider what a trend claim must contain to count as research.

(i) The regularity: what has been changing, measured how, and since when.

(ii) The reasons it holds: a mechanism, or the inertia of slow structures (Section 4).

(iii) The span: for how long those reasons can be expected to hold.

A megatrend list usually offers the first item, a name for the second, and nothing for the third. Without them, "megatrend" is a big name for "things are changing", and the list works as if it were a view of destinations that the trends were only ever directions towards. What would break a trend is a different question, and a research question of its own (Section 6).

Someone might reply that megatrend lists are orientation devices, not research, and that nobody takes them as findings. If so, I have no quarrel with them. The quarrel begins when

they are cited as evidence in research, as grounds for a picture of 2040, and so on. Then the marketing department has been promoted to the methods section.

4. What holds, and for how long

If knowledge of futures comes from what we know about how things work and about the present situation, two questions follow. Where does futures studies get its knowledge of how things work? And what entitles us to think that this knowledge will keep holding? This section takes up the first question in (A) and the second in (B) and (C). It then turns to two common sources of claims about futures: experts and numbers in (D), and reports built on other reports in (E).

(A) *Borrowed knowledge*. Most of what futures studies knows about how things work, it borrows: from demography, economics, climate science, sociology, engineering, and so on. This is nothing to be ashamed of. Individual sciences cannot map futures on their own, because scientists disagree, regularities from different domains have to be combined, and the present situation is contested (Virmajoki, 2026a). But the knowledge that goes into any combination comes from them. Futures studies is, in this sense, a borrower, and a good borrower knows exactly what was borrowed, from whom, and on what terms.

A bad borrower does something else. The field can borrow and combine knowledge, and it should, since combining is where it can add something of its own. But it must be extremely careful that what it offers is not merely a repackaging of what is already known elsewhere. Megatrend lists are the clearest risk (Section 3): they sell well, but they often add little to what the sciences they draw on already know.

There is a recurring demand that the field should have theories of its own. Fergnani and Chermack (2021) attribute the field's failure to win a place among the established social sciences to its resistance to scientific theory and, drawing on management and organisation science, set out what counts as scientific theory in the study of organisations. Theory of foresight practices in organisations is a fine thing to have, and testing what foresight does is research (Section 7). However, a demand for theory has to say what theories are and how they are tested. That question is old. Philosophy of science spent much of the twentieth century on it: theories as axiomatised sets of statements tested against observation (Nagel, 1961; Hempel, 1965), falsification (Popper, 1959), paradigms (Kuhn, 1962), research programmes (Lakatos, 1970), and finally what reads as an obituary for the view the century began with (Suppe, 1977). The years tell their own story. A proposal that does not say where it stands in this history risks taking over, implicitly, its oldest and most criticised view, which would be worse for the argument than taking none. Declaring that this is what the field must do makes the field narrower, not stronger.

Worse is the idea of a “theory of the future”. Theories are about things: cells, markets, institutions, and the ways they work. The future is not a thing in that sense, and a theory of it is a category mistake (Ryle, 1949). What futures studies can have theories about are present things with modal features (structures, tendencies, the possibility space of a domain; Virmajoki, 2026a), and those theories, too, are mostly borrowed.

(B) *Continuation*. Every claim about futures assumes that something keeps holding. The age structure keeps ageing, the institutions keep working, the regularity keeps regulating. This is unavoidable, and it is no scandal. Hume showed that no experience of the past can by itself

guarantee the future, and that we live by such expectations anyway: “Custom, then, is the great guide of human life” (Hume, 1748/2007, Section V, Part I). Hume was no radical sceptic, though he is often read as one. What matters here is another of his points: we never perceive the connection between cause and effect, we infer it. The future, likewise, is reached by inference, never by sight.

If the field adopted only one duty specific to futures research, it should be this: state what is assumed to keep holding, and why. Two kinds of reason do most of the work.

(i) Mechanism. We know why something holds, and nothing in sight disturbs the mechanism.

(ii) Inertia. The structures involved change slowly, as infrastructure, institutions and age cohorts do. Their time-needs are long.

Neither reason refutes Hume. Reasons make continuation assumptions better, not certain. But a stated assumption with its reasons is a premise that others can inspect and reject. An unstated one works as if it were the telescope’s mount: it holds the picture steady while nobody looks at it.

(C) *Two kinds of uncertainty*. Scenario axes, and much else in futures studies, are justified by uncertainty. But “uncertain” covers two different things, and we must separate uncertainty that stems from the following.

(i) Ignorance. We simply do not know. The answer could be found, and better methods would remove the uncertainty. The remedy is research.

(ii) Contingency. The world has not yet decided. Decisions are still to be made, and events may or may not happen. No method, however sure, removes this. Here we let the world decide, and an axis, a branch or an alternative is the honest way of saying so.

Contingencies of type (ii) cannot be known for sure by anyone in time. Early religious writers, Augustine among them, imagined a god for whom all times are present at once (Augustine, *Confessions*, Book XI). Such a god would be the only candidate owner of a metaphysical telescope. But the god they talked about has nothing to do with us as researchers. An axis should mark what the world has not yet decided, not what we have not yet bothered to find out.

We also need to separate ignorance as a topic from ignorance as our predicament. Actors are often ignorant. Decision-makers may not know what the ground along a planned railway will carry, and whether they find out in time is part of how things will go. Their ignorance is a feature of the situation, and it may well be a dimension of a scenario set. Our own ignorance as researchers is different. It belongs in the limits section (Section 8), flagged as ignorance. Sometimes, when finding out would take longer than the decision allows, it may enter a scenario set too, but flagged, never dressed up as contingency. A parent may not know where the child is, and the child may not know where Jupiter is. The first is the parent's predicament; the second is a fact about the child.

The distinction is not always sharp. The literature on decision-making under deep uncertainty defines its subject as the condition in which analysts do not know, or parties cannot agree on, the appropriate models, the probabilities, or how to value outcomes (Lempert, Popper &

Bankes, 2003). That is an honest mixture of (i) and (ii), and it is fine as long as it is declared as such. What is not fine is ignorance dressed up as contingency: an axis, say, for something that a week in the library would settle.

(D) *Experts and numbers*. Where knowledge of how things work is thin, the field turns to experts. The Delphi method was designed for exactly this (Dalkey & Helmer, 1963): ask experts, feed back the distribution of their answers, ask again, and watch the answers converge. What a Delphi study delivers, however, is research data about the present: what a panel of people believes now, and how far they agree. That is worth knowing. It is evidence about futures only indirectly, through whatever the experts know about how things work and about the situation we are in. A consensus is not a sharper picture assembled from many telescopes. It is a fact about a panel. The better versions of the method know this, and ask experts for their arguments rather than only their numbers, so that the arguments can be examined (Kuusi, 1999).

The same holds for surveys of expert opinion. A large survey of AI researchers put the year by which high-level machine intelligence has an even chance of arriving at 2047 (Grace et al., 2024). The figure is data about AI researchers: what they believe, and how their beliefs are distributed. Averaging beliefs yields an average of beliefs. 2047 is the date of a calculation, not of an event.

Behaviour, too, is evidence. What people bet on, buy and plan for tells us what they expect, and markets for predictions can aggregate such expectations well (Wolfers & Zitzewitz, 2004). If nearly everyone who bets on football expects a club not to win the league, that is

evidence about how the season will go, and it would be odd to ignore it. I am not a sceptic about evidence. I am sceptical about sightings.

Forecasting can also be studied as a skill. Tetlock and Gardner (2015) found people who forecast better than others, and they are clear that this is no mysterious gift one is born with: it can be learned, practised and improved. They report that the average expert's accuracy fell sharply as the horizon lengthened, approaching chance three to five years out; they take this as evidence of limits that may constrain even superforecasters. Both points matter here. There are no born seers, and no trained ones either beyond the reach of what they know. Futures studies researchers are made by training, not born with an eye for 2050. Dylan gives the point a fitting image in "Desolation Row": the moon and stars withdraw from view, and even the fortune-teller packs up and goes inside (Dylan, 1965).

(E) *Reports built on reports*. Consider an imagined but familiar document: a report on the future of some sector, forty pages long, with confident claims about 2040 and four references, two of them other reports of the same kind. The number of sources is not the problem; it is a symptom. If four sources will do for claims about 2040, why forty? And if forty will do, and four, why not none? Claims about futures can rest on few sources only if the sources are of the right kind and the claims are drawn from them with their conditions attached. A projection from a statistical office, or an assessment of the climate system, can carry a great deal if it is read through and its assumptions are carried along. Too often neither happens. The references are chosen to decorate claims that were made first, and made as if by sight.

Cassam (2018) has a name for the attitude behind such work: epistemic insouciance, a casual indifference to whether one's claims are grounded in evidence. He links its typical product

with what Frankfurt (2005) analysed under a less polite name. Generative AI has now industrialised the genre: the four-source report can be had with four million sources, none of them read. As a tool, it is as good as the people who make its assumptions explicit and check them.

I put the blame on structure before individuals. Shortcuts are rewarded. Decision-makers ask for pictures of 2040, and pictures are funded and sold. The field teaches the cone and the disclaimer (“we do not predict”) in its first lessons, and the disclaimer then does the work of method. Research is traceable, open to scrutiny and transparent about its assumptions, and it does not happen once; it is checked and redone. The individual report is an exhibit of the system at work, which is why I name influential texts and imagine ordinary ones.

5. Possible, plausible, probable

(A) *The modal vocabulary.* Futures studies speaks a modal language: possible, plausible, probable, preferable, and so on. The words are used as if they were easy. They are not. Modal logic, metaphysics and epistemology are among the hardest parts of philosophy, and those who shaped them did not find them easy either (Hintikka, 1962; Kripke, 1963; Plantinga, 1974). I do not offer a theory of modality here, which would be exactly the kind of hubris this paper argues against. I offer one distinction, and I insist on it. We must keep two kinds of modality apart (for the varieties of modality, see Kment, 2021).

(i) Objective modalities. These concern the world: what is there, how things work, and so what can still come of the present situation, and how strongly things tend one way rather than another. A weather forecast for tomorrow reports, in its probabilities, what the patterns of the

atmosphere show. For futures, the central notion is historical possibility: what remains open, given what is already settled (Belnap, Perloff & Xu, 2001).

(ii) Subjective modalities. These concern us: what we believe, how much we trust it, and what we want. They are about our minds and our values, not about the openness of the world.

The subjective side splits. Part of it is cognitive: what we believe, and how strongly.

Evidence runs through this part: what evidence we have, and how far our degree of belief answers to it. “A wise man, therefore, proportions his belief to the evidence” (Hume, 1748/2007, Section X). Degrees of belief can be disciplined further: they must at least be coherent, and they can be tested by bets (Ramsey, 1926/1931; de Finetti, 1937). The other part is evaluative: what we would prefer. Wishes answer to values, which can be stated and argued for. Whether a wish can come true, however, is again a question about the world.

Of course, every modal claim, of either kind, is made by humans with limited knowledge.

But a claim made by fallible humans can still be about the world. When a weather forecaster says that there is a 70 per cent chance of rain tomorrow, the forecaster may be wrong. The claim, however, is about the atmosphere, and it is checked against the atmosphere: on the days for which 70 per cent was said, did it rain about seven times in ten? When I say that, for all I know, it may rain tomorrow, the claim is about my knowledge, not about the sky.

Perhaps I simply have not looked at the forecast. The two kinds of claim can come apart, and in both directions. For all I know, my old car will pass next year’s inspection; given the rust underneath, it will not. Then again, a town that has not flooded for a hundred years may take its records to rule out a flood, while the river and the rain have ruled out nothing. Kripke (1980) made the point with an example that goes back to Frege (1892). For ancient

astronomers, it was possible, for all they knew, that the Evening Star and the Morning Star were two different bodies. It was not possible at all: both are Venus, and nothing can be distinct from itself. The objective modality that matters for futures is historical rather than metaphysical, but the lesson carries over unchanged.

(B) *The cone*. Consider the futures cone, one of the best-known pictures in the field. From the present, a cone opens towards the future, and inside it lie zones: the probable inside the plausible, the plausible inside the possible, the preferable cutting across them, and in some versions a band of the preposterous outside the possible. The picture has a long history. The sorting of futures into possible, plausible, probable and preferable goes back at least to Henchey (1978), and the cone to Taylor (1990) and Hancock and Bezold (1994); Gall, Vallet and Yannou (2022) trace what followed. Its best-known form is due to Voros (2003, 2017). My quarrel is not with any one drawing but with how the picture is used.

In practice, the cone is filled in a workshop, and the filling sounds like this: “I feel this is plausible but not probable.” Feelings go in, a picture comes out, and the picture is shown back to the room as if it were a view of futures. The worse problem is what the picture forgets. Every word on the cone except “preferable” has an objective reading and a subjective one. “Probable” may be the forecaster’s 70 per cent or my hunch; “possible” may be what the world leaves open or what I cannot rule out; “plausible” may be a pathway from the present situation or a feeling in the room. Once an idea sits in a zone, nothing on the picture says which reading put it there, or on what grounds. Every zone looks the same, however it was filled. Trust, as we saw, can be disciplined by coherence, by bets, by scores. The cone asks for none of this.

The picture has troubles of its own. Zones can be nested only if they answer to one measure, and these do not: the possible answers to knowledge we do not yet have, the plausible to current understanding, the probable to current trends, the preferable to values (Voros, 2017). Knowledge closes doors as well as opening them. Inventors once took perpetual motion machines to be buildable, and thermodynamics showed that they are not. The plausible of one day is not always inside the possible of the next. The preposterous band adds a stranger twist. It holds futures judged “ridiculous” or “impossible”, and its motive is sound: some ideas once laughed at came true. But impossible is impossible. A future that comes true was never impossible; it was misjudged. Drawn as a band outside the possible, a misjudgement of ours is placed in the world as if it were a place.

Finally, the cone is an empty picture. With no words in it at all, it still seems to say something, and it works as if it were a Rorschach inkblot: what one sees in it tells about the one who looks. Its author’s own gloss says as much, since the zones are “subjective judgements ... based in the present moment” (Voros, 2017). An inkblot is a fine instrument for studying the people who look at it. It shows nothing of futures.

The cone began as a simple device for sorting futures. It has since got out of hand, in two ways. It has multiplied: it has been redrawn, extended and revised many times, with zones and labels added along the way, and the review that traces its history proposes yet another revision (Gall, Vallet & Yannou, 2022). One added band, the preposterous, was drawn in homage to Dator’s Second Law: any useful idea about the futures should appear ridiculous (Dator, 2009b; Voros, 2017). Openness is a virtue in researchers. Made into a law, it draws a zone for the ridiculous. No revision that I know of asks what a zone answers to. And its mission has crept. A device for sorting futures in a workshop is now used as if it were a

research instrument, and what is placed in its zones is reported as if it were a result. I do not offer a better cone. The trouble is not in the drawing, and a better drawing would only give the inkblot a finer frame.

(C) *Plausibility*. Of all the words on the cone, “plausible” carries the most weight in scenario work. Scenario literature has often distinguished plausibility from probability; Ramírez and Selin (2014) survey this opposition and argue that the either/or distinction should be surpassed. What, then, is plausibility? In practice, it is a judgement made in the room. A scenario is plausible if the participants, or the client, find it so, and the literature has been open about this: plausibility has been described as something negotiated (Selin, 2011). Negotiation yields agreement. It does not yield grounds, and a scenario that the room finds plausible may be plausible, or it may merely be familiar. Plausibility needs grounds, not a vote.

Properly understood, plausibility belongs to the objective side, and like everything there it is assessed fallibly. A future is plausible to the extent that there is a pathway to it from the present situation, given what we know about how things work: few and small departures from how things now go, each of them something that can happen. The closer a future lies to the course things are actually taking, the more plausible it is. Plausibility in this sense can be felt, but it is not a feeling. It is a feature of the pathway, and a claim about it can be checked by laying the pathway out, step by step, so that others can see where it is weak. This also explains why plausibility and probability travel together, roughly: what lies close to the actual course tends to be more probable than what lies far from it. I do not offer a theory of plausibility. I only insist that “plausible” should mean something that can be argued for and against, and a room’s agreement is not such an argument.

(D) *No future, no error*. There is another way out of the modal mess: deny that there is anything to be right or wrong about. One version says that the future cannot be predicted because it does not exist. Futures literacy, in UNESCO's words, starts from the claim that "the future does not yet exist, it can only be imagined" (UNESCO, n.d.). Part of this I accept: the many futures of futures studies are not out there (Section 1). But the conclusions do not follow, and the claim can be read in two ways.

(i) The trivial reading. Future things are not present. Of course they are not; that is what "future" means. Whether they are, in some other sense, really there is a question for metaphysics. I sit on a chair, not on the sofa, and the sofa is no less there for that; whether times are like places in this respect, I leave to metaphysicians. Nothing in futures studies depends on settling the matter, and nothing about prediction follows from it. The total solar eclipse of 12 August 2045 is not present either, and it is predicted to the minute. The prediction needs the present positions of the Sun, the Moon and the Earth, and how the solar system works. It needs no future object. From "not yet" nothing follows about "only imagined": the eclipse is not yet, and it is predicted, not imagined.

(ii) The heavy reading. There are no facts about the future at all; nothing about it is yet true or false. This is a serious metaphysical thesis, debated since Aristotle's sea battle (Aristotle, *De Interpretatione*, 9), and the field asserts it without arguing for it. Even if it were true, it would cover only what is still open. What the present situation has already settled can still be predicted: the eclipse, or the number of people over 80 in 2045. The familiar defence that only social futures are open is reading (ii) restricted to society, and the same answer applies.

Either way, the conclusions do not follow. The claim takes its truth from the first reading and its punch from the second, and it slides between them. The slide does a convenient job. If the future does not exist, or can only be imagined, nothing said about it can be wrong, and the field is immunised against error. Nothing remains at the end of the telescope but the lens.

The slide also leaves the field unsure whether it has anything to study. Futures literacy is meant to help people understand “the role of the future in what they see and do” (UNESCO, n.d.). If the future can only be imagined, this is the role of imaginings, and then one must ask what makes one imagining better than another. The answer cannot be another imagining. The usual reply is a gotcha: then you can only study the present. Yes, and that is enough. The present contains what shapes what can come of it: structures, tendencies, plans and decisions, and the ways they bear on one another. Studying these is studying the possibility space of a situation (Virmajoki, 2026a), whether or not futures are, in the end, really there. That needs no metaphysics. It needs evidence.

6. Scenarios, surprises and pictures

(A) *Sets, not pictures*. Scenarios are the field’s signature product, and the word covers almost anything: a narrative of 2040, a model run, a workshop’s poster. I do not try to say what all of them are; the practice is too varied to have one essence. I suggest instead what the best of them are, and so what the word should mean in research.

The classic definition is already more demanding than much of the practice. Kahn and Wiener (1967) defined scenarios as hypothetical sequences of events, built to focus attention on causal processes and decision points, and they asked of them, among other things, what

alternatives each actor has at each step. Each scenario is one sequence. Its point is plural: a decision point is a place where things can go another way.

My suggestion is this. A scenario, in research, is a member of a structured set: descriptions of possible situations that vary along stated dimensions, the things taken to be able to vary, and that agree on what is held fixed. The familiar two-by-two matrix is the simplest case. Its two axes say: if we move here, this is what we get. The axes are too often simplistic, but the form is right, since it makes explicit what is taken to vary, and so what is not. A single description of a possible situation is not, on this suggestion, a scenario. There is nothing wrong with it; it should simply be called what it is. A scenario alone is a picture. A bundle is an argument.

Not every set needs axes in the strict sense. Some scenario sets are clusters, built from many considerations. Even then, the members must share variance in some stated dimensions, or the set is no set, only a handful of pictures, each written as if seen from somewhere. There are no alternatives from nowhere.

The dimensions must be stated, and argued for. Why these and not others? Why are they taken to vary, and over what span? Here the distinction of Section 4 returns: a dimension should mark what the world has not yet decided, not what we have not yet bothered to find out. Any choice of dimensions is allowed if it is argued for; no choice may be hidden.

Some methods already work this way. General morphological analysis lays out the dimensions of a problem and their values, and then asks which combinations hang together (Ritchey, 2011). Exploratory modelling runs a model across many combinations of uncertain inputs and asks which outcomes depend on what (Bryant & Lempert, 2010; Kwakkel &

Pruyt, 2013). Neither settles anything once and for all; both open new questions, as research does. But both say what varies, and both can be checked.

(B) *Prefabricated futures*. If a scenario set must vary along dimensions drawn from the situation at hand, a set that comes ready-made, before the situation is even known, is suspect from the start. The field has several. Dator's four generic futures, Continued Growth, Collapse, Discipline and Transformation, are offered as types into which images of the future fall, and as a frame for building scenarios about anything (Dator, 2009a). Fergnani and Song (2020) derive six scenario archetypes from science fiction films set in the future, and offer them to scenario practice.

Inspiration can come from anywhere, films included. Where an idea comes from is a matter of discovery, not of justification, a distinction I return to in Section 7. But a typology offered for research has to be shown to be a good one: that its types are distinct, that together they cover what they claim to cover, and that they fit the job they are offered for. That the plots of films, or the images held in a culture, fall into a few types is not yet a reason to think that the futures of a domain do. The burden is on those who offer the types, and evidence about films or images does not discharge it.

The trouble grows when the types are used as scenario sets. Then the dimensions along which the futures of a domain are taken to vary are not drawn from the domain at all. They come from a culture's stock of stories, or from the cinema, and they are applied to energy, health care and universities alike. A frame that fits every situation fits none in particular. It is as if the telescope came with four slides already in it, and every view had to be one of them.

Prefabricated futures are alternatives from everywhere, which is to say, from nowhere. They save exactly the work a scenario set exists to do: finding out what, in this situation, the world has not yet decided. A workshop that begins with the four generic futures has answered its first question before asking it.

(C) *The shovel in the living room*. Wild cards are events of low probability and high impact, and lists of them are a staple of foresight (Petersen, 1999; Mendonça et al., 2004). A wild card, however, is never wild on its own. A child with a shovel in a park is not wild. The same child with the same shovel on the living-room floor is. Wildness is a relation between an event and a picture of how things normally go, and a list of wild cards depends on such a picture, as a parasite depends on its host. Usually the picture is not stated. The cards are wild against a view of 2040 that nobody has argued for, as if it had been seen.

The honest version states the picture. Kahn and Wiener (1967) did this with their “surprise-free projection”, an explicit account of what is assumed to go on without surprise, so that a surprise can be defined against it. That is the living room, described, and the shovel can then be placed with some sense. Better still is to turn the question around. Instead of listing what might fall from the sky, ask where the present situation is fragile: which of the things assumed to keep holding (Section 4) could stop holding, and what would follow if they did. Where a trend might break is a research question of its own, different from where the trend is going, and it needs its own evidence.

None of this removes the deeper limit. We cannot conceive of everything. The history of science is full of alternatives that nobody conceived at the time, though the evidence then

available would have supported them as well (Stanford, 2006). Futures are no different. This is a reason for humility, not a method.

(D) *Ignorance as method*. Scepticism can claim too much as well. On one influential view, what matters most in history comes as a Black Swan: rare, of extreme impact, and explained only after the fact, and so the course of history cannot be predicted (Taleb, 2007). Notice what this claims. It does not say that some forecasts lack grounds, which is often true. It says what the future will be like: dominated by what nobody can foresee. That is a claim about all futures, made without the grounds it denies to everyone else. The sceptic, too, claims a telescope. It is a special one, as if fitted with a lens from a metaphysics shop on Penny Lane, and through it one sees that the moon is not there.

Kripke's remark has a first half that I have so far left aside: possible worlds, he said, are stipulated (Kripke, 1980, p. 44). Futures are not ours to stipulate. The extremes stand the remark on its head. The confident forecaster and the radical sceptic both stipulate what is and is not possible in futures that nobody can know that well, and both report the stipulation as if it were a sighting: I see this and that, so this is possible and that is not.

Ignorance can also be made into a procedure. Building on Taleb's notion of antifragility, Derbyshire and Wright (2014) develop a methodology that complements scenario planning "by omitting causation". Their honesty about the limits of causal knowledge deserves respect. But honesty earns tolerance, not the title of research. Causation, as Section 2 argued, is what connects the present to what can come of it, and a method that omits it omits the only bridge there is. Someone might reply that decision-makers need preparation, not causes, and that

serving them is why the field exists. That may be so, but it tells us nothing about research. A good salesperson can sell fortune-telling to decision-makers. It is still no research.

(E) *Pictures and stories*. The field is keener on drawing than on thinking. Futures work comes with matrices, radars, cones, and illustrations of 2040 with clean streets and quiet drones. Pictures are not the problem as such. A diagram can be research when its shape carries an argument. The double helix is the classic case: its structure showed at once how the molecule could be copied, as its discoverers noted (Watson & Crick, 1953). The picture encoded what depends on what. Most futures pictures encode nothing of the kind. Their arrows do not say what relation they stand for, and their boxes are not variables. That good diagrams exist elsewhere does not help, as if a thousand masterpieces in the Louvre could make the stick figures in my drawer into Picassos.

Stories are a different matter, and a more serious one. Scenarios have long been told as stories, and the classic manual recommends them because they move people in a way that graphs and equations do not (Schwartz, 1991). Graphs and equations do not move people that way; they have to make do with carrying an argument. A story is a good format: it carries a sequence, it can make a set of possibilities understandable, and narrative foresight can be done with method (Heinonen, Virmajoki & Viitamäki, 2026). But the impact is exactly what should worry us. A story persuades by detail and coherence, and detail misleads in a known direction. Tversky and Kahneman (1983) asked 245 University of British Columbia undergraduates how probable two events were (each half of the sample rated one): a massive flood somewhere in North America in which more than a thousand people drown, and an earthquake in California causing a flood in which more than a thousand people drown. The second was judged more probable, though it cannot be, since it is a special case of the first.

Each added detail makes a story feel more plausible and makes it less probable. The impact that helps a story communicate is, as evidence, a bias. It is the felt plausibility of Section 5 (C), now documented.

Pictures and stories can carry an argument. They cannot replace one.

7. Wishes, workshops and imagination

Not all futures work describes what can come. Much of it describes what should come, and then asks how to get there. This is legitimate, and much of it is valuable. It is also where the as-if telescope hides best, because a wished-for world is easily treated as a destination waiting out there, as if it had only to be reached.

(A) *Metaphysical tickets*. Section 2 showed that Gibson's plane ticket takes you somewhere else, not somewhen else. Backcasting sells a ticket of another kind: a return ticket from 2050. The method was proposed with care. Robinson (1982) introduced it in energy policy as a way of asking how a desirable goal could be reached and what reaching it would require, as an alternative to asking which futures are likely. Dreborg (1996), for example, likewise presents backcasting as a problem-solving and teleological approach. In much later practice, however, the end point is no longer a goal but a described world: a vision of 2050, rich in detail, from which the participants then travel back, step by step, to the present. The trip back is the selling point. But a return ticket presupposes an outbound trip, and nobody has made it.

To describe the end point, one must assume a great deal about 2050: not only technologies and institutions, but values, meaning what people will want, fear and accept. Values vary

across time and may differ substantially in the future (Danaher, 2021). That makes it risky to assume that people in 2050 will share today's values. A vision of 2050 written in today's values is a picture of today, dated 2050. Even a bare target assumes more than it seems to. Take an imagined one: half of all trips in a city made by bicycle by 2045. It looks like a number, but a pathway back from it takes for granted the world in which the number is reached: where people live and work, what else moves them about, and whether they still want what the target assumes they want.

There is an honest version, and it comes from shrinking. Take a target modest in size, a span within the reach of what we know (Section 3), and actors who control the steps. Then working backwards is planning, and planning is respectable. Such a plan is built as if it were a bridge from the near bank, on what we know about how things work, and it can be checked as it is built. What it cannot do is start from the far bank.

The premortem may look like a ticket, but it is not one. Imagining that a project has already failed, and asking why, is a device for thinking better now. People find more reasons for an outcome when they imagine that it has already happened (Mitchell, Russo & Pennington, 1989), and teams use the device to bring out the risks in their plans (Klein, 2007). No view of the future is claimed. The failure is imagined in order to examine the plan, and the plan is present. That is the difference between a tool for thinking and a ticket for travelling.

(B) *Seeds and visions*. I have called a new practice that starts somewhere a seed, not a sighting (Section 2). A research programme uses the same word. The seeds of good Anthropocenes are present initiatives, practices and technologies, marginal now, that could contribute to better futures if they spread (Bennett et al., 2016). Up to this point we agree.

Seeds are present things, and what they need in order to spread, and what would stop them, can be studied.

The trouble lies in the next step. In workshops, seeds are combined into visions: rich descriptions of good futures in which the seeds have grown and joined (Pereira et al., 2018). From a seed one can say, with evidence, what it needs and what threatens it. A vision describes the grown garden. It answers to the wishes and the imagination of the people in the room, not to what holds, and it is then reported as if it had been seen. As a way of motivating people and building coalitions, it may work well. That is practice. As a finding, it is a picture of what the room hopes for.

The field's trust in visions has a founding story. Polak (1961/1973) writes that “the rise and fall of images of the future precedes or accompanies the rise and fall of cultures” (p. 19), and elsewhere gives such images an active social role. That association is repeated far more often than it is tested. In its weak form, that images of the future can act on what happens, it is trivial: plans and expectations shape what people do, as Section 2 already noted, and nobody needed Polak to say so. In the strong form often drawn from it (that a civilisation rises and falls with its images of the future), it is a founding myth. Repeating it is not testing it.

(C) *Imagination and discovery*. Imagination has a proper place in research, and an honoured one. Philosophers of science long ago separated the context of discovery, where ideas come from, from the context of justification, where they are tested (Reichenbach, 1938). Ideas may come from anywhere. Kekulé said that the ring structure of benzene came to him in a reverie of a snake seizing its own tail (Benfey, 1958). Chemistry did not ask where the idea came

from, only whether it held. Futures studies needs imagination for the same reason every science does: nobody can test an alternative that nobody has conceived (Section 6 (C)).

Two jobs follow, and they must not be confused. One is to conceive: to widen the range of what is considered. The other is to ground: to say, of what has been conceived, what could come of the present situation, on what grounds, and what can be ruled out. Methods of imagination, such as design fiction and experiential futures (Candy & Dunagan, 2017), are tools for the first job. What they deliver are hypotheses, not results. Publishing their outputs as findings confuses discovery with justification. Futures literacy laboratories go one step further. They are presented not only as exercises in imagination but as a way of creating knowledge (Miller, 2018), so the confusion is built into the design. Imagining together is still imagining. The field has built an industry around discovery and left justification half empty.

Dator's Second Law says that any useful idea about the futures should appear ridiculous (Dator, 2009b). Dator immediately concedes the obvious problem: many ridiculous ideas remain ridiculous. That is the point. Ridiculousness may protect discovery from premature dismissal; it justifies nothing. Looking ridiculous proves nothing, either way. A list of ideas once laughed at that came true says nothing about the odds. Wildness may help discovery. It never justifies anything.

Not everything that sparks an idea is research. Tarot cards can prompt a new thought, and so can a novel. But tarot answers to nothing in the world, and a novel answers to its story. Both may serve discovery; neither can serve justification, because neither is attached to how things are. Futures studies can have many methods, but not anything goes.

(D) *Workshops*. Much of futures work happens in workshops, and the workshop has become the field's signature event. What does a workshop produce? It can produce three things of value.

(i) Knowledge. The participants know things about how their domain works and where it is heading, and a workshop can bring this out, much as a Delphi round does (Section 4).

(ii) Seeds. New ideas surface, and some of them deserve study.

(iii) Legitimacy. People who have taken part are more likely to accept what comes out, and to act on it.

None of these is a research result. The knowledge is data, to be checked like any other; the seeds are hypotheses; legitimacy is a social good, not an epistemic one. A workshop is practice. Holding one is not the same as having done the research.

The practice can, however, be studied honestly, and some of the best work in the field does exactly that. Scenario workshops are interventions in the present, and their effects can be measured. In experiments, scenario planning has been found to reduce framing bias in strategic decisions (Meissner & Wulf, 2013). In the field, it has changed how experts judge long-range investments (Phadnis et al., 2015). This is research in the plain sense: a claim about what a practice does, tested against evidence. It needs no view of 2040, only the present and its people.

Generative AI can now fill a wall of sticky notes in seconds. Discovery has never been cheaper. Justification costs what it always did.

8. What the field is when it is honest with itself

(A) *An old honesty*. Nothing in this paper would have surprised the field's founders. They knew what they could and could not get, and they said so in their very titles. Helmer and Rescher (1959) wrote on the epistemology of the inexact sciences. De Jouvenel (1964/1967) called his book *The Art of Conjecture*. Kahn and Wiener (1967) subtitled theirs *A Framework for Speculation*. Inexact, conjecture, speculation: these are not the words of people who thought they had a telescope.

None of this is to say that the field lacks good research. It does not. My claim is narrower: the honesty the founders put into their very titles has, in many corners, given way to a tone that promises more than the work delivers.

Attempts to give the field foundations have been made. Amara (1981) sorted the field's tasks into the study of the possible, the probable and the preferable. Bell (1997) tried literally: two volumes called *Foundations of Futures Studies*. As philosophy, they are an amateur's collection of thoughts, sold as foundations. Their key concepts are listed far more often than they are opened up. They are also the closest the field has come to foundations, and that tells its own story. Poli (2017) has tried again since, from anticipation. The field, on the whole, preferred pictures.

Compared with philosophy, which can be so strict that it goes nowhere, futures studies has room to breathe. It has too much. Room is good for discovery. Justification needs walls.

(B) *A limits section.* If one practical change could be made tomorrow, it would be this. Every study that makes claims about futures should include a limits section, and editors could require it. It would state three things.

(i) Grounds and spans. What is assumed to keep holding, why, and for how long (Section 4).

(ii) Provenance. Who assembled the claims, from which inputs, and how disagreements were settled.

(iii) Dimensions. What was taken to vary, and why, and whether each dimension marks what the world has not yet decided or what the authors did not find out (Sections 4 and 6).

A limits section costs a page. It turns pictures back into claims, and claims can be checked. A study of 2050 built in a workshop could pass review with such a section; without one, it should not. Empirical fields have long had their limitations paragraphs, and nobody there takes one for a confession of failure.

(C) *Assembled, not viewed.* What futures studies offers, at its best, is assembled. Knowledge of how things work is borrowed from the sciences (Section 4). Evidence about the present situation is gathered, and contested (Section 3). Dimensions of variation are chosen and argued for (Section 6). Out of these, researchers assemble an account of what can come of the present situation, its possibility space (Virmajoki, 2026a). This is real work, and it can be

good work. But it is assembled, not viewed, and the as-if telescope hides exactly that. A view has no author; an assembly has one. Somebody chose which knowledge to borrow, which present to start from, which dimensions to vary and how far to look, and those choices serve somebody's needs: whose time-needs set the horizon (Virmajoki & Laakkonen, 2024), whose questions were asked, whose disagreements were settled, and how. Presented as a view, the assembly hides its assemblers, and with them the power that went into it.

(D) *What the field is.* Futures studies often introduces itself by what it does not do: it does not predict. Then it shows pictures of 2040. A field that defines itself by a denial and then breaks it makes a joke of itself. It would do better to say what it does. Here is my suggestion, in two sentences. Futures studies makes explicit, and tests, the assumptions behind claims about futures. It says what can come of the present situation, on what grounds, for how long, and where knowing stops.

This is not a demotion. It is what the best work has always done. Wack's scenarios at Shell worked because Wack knew the oil business thoroughly, and because he aimed them at the managers' present perceptions, not at a picture of the future (Wack, 1985a, 1985b). Turing (1950) ventured a claim about machines some fifty years ahead, with a measure, a span, and grounds in what he understood about computing machinery. It could be checked, and it was partly right. Neither looked through a telescope. Both understood something, and said modestly what followed from it.

Possible worlds, Kripke said, are not discovered by powerful telescopes. Neither are futures. If you want to be a researcher, be honest with yourself. There are no metaphysical telescopes.

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